Abstract

The proliferation of digital services has led to users often validating Terms of Service (ToS) and privacy agreements that are lengthy, complicated and hard to understand in their entirety without legal skills. As a result, users may voluntarily agree to clauses that affect their privacy, financial responsibilities, and legal rights. In order to meet this challenge, this project introduces RiskWise, which introduces a web-based intelligent system that can automatically analyze ToS documents and also emphasizes potentially risky contractual clauses in an easy to comprehend and explainable way. The proposed architecture combines Natural Language Processing, Knowledge Graph generation and Retrieval-Augmented Generation in a single hybrid model.

Uploaded documents are semantically parsed and chunked and then embedded into document embedding in a pattern and similarity-based retrieval. Concurrently, structured subject-predicate-object relationships are taken to build a knowledge graph that reflects contractual entities and obligations.

A hybrid retrieval framework built upon vector similarity search and graph traversal extracts enriched contextual information, which is subsequently analyzed by a generative language model in order to detect risky clauses, grade them for severity and generate user-friendly explanations. The functional prototype was created with a modular full stack model using a React frontend, FastAPI backend, and graph database that supports relational and vector versions.

Case-based experimental testing validated the capability of the system to retrieve contextually relevant clauses, justify their nature, and allow interactive user exploration via conversational querying.

These findings show that hybrid Graph-RAG architectures can indeed be applied to facilitate contextual completeness and interpretability of automated legal document analysis. RiskWise bridges these gaps by translating complex contractual content into actionable insights, helping provide transparency for the digital experience, supporting informed consent, and educating the audience of users on what they can do in a digital landscape.

Keywords: Terms of Service Analysis, Knowledge Graph, Retrieval-Augmented Generation, Legal NLP, Hybrid Retrieval, Explainable AI

Chapter 1 — INTRODUCTION

Digital services have been growing exponentially and changed the way people are approaching technologies, allowing for the easy access to any apps, platforms, and the whole digital ecosystem. This expansion, however, has also led to an increased dependence on digital contractual agreements, such as Terms of Service (ToS), Privacy Policies, and End-User License Agreements (EULAs) which users have to accept prior to accessing services. Although these agreements provide critical legal foundations from which users and service providers can access and trust the service, their long-form and convoluted nature may make them difficult to read critically and understand, and pose issues regarding informed consent and digital transparency. However, the recent development of AI and Natural Language Processing is enabling automated analysis of challenging textual information with no prior understanding. Techniques like semantic embeddings, knowledge graphs, and Retrieval-Augmented Generation can help smart systems obtain insights out of unstructured documents whilst preserving the relationships in meaning. Utilizing these advancements into an AI based framework to analyze legal agreements, the RiskWise initiative recommends the generation of intelligent risk-sensitive insights when it comes to understanding legal and other documents of the public domain. In this chapter, the background, motivation, and problem formulation of the proposed system is presented.

1.1 Context and Background Study

As the scope of digital services and online platforms continues to grow rapidly, users are becoming more and more tied into Terms of Service (ToS), Privacy Policies and End-User License Agreements (EULAs) for gaining access to apps/services. These documents act as legally binding agreements between service providers and users detailing rights, obligations, limitations of liability, data usage policies, termination conditions, or other contractual provisions concerning service usage. Such agreements are absolutely necessary in order to make legal relationships between parties in digital ecosystems, but as the general user experience is often limited, a lot of the practical effectiveness is undermined. While significant, empirical studies and user behavior data routinely demonstrate that most users either very rarely read these documents in their entirety, or do not fully understand them. The top reasons are: too lengthy documents, complicated legal language, intricate structure styling and extensive cognitive labor, especially to understand. This general disengagement results in a great divide between user informed consent and user understanding, that has implications for the legitimacy of informed consent in the digital realm. Thus, users are very frequently accepting agreements without knowing provisions that allow for comprehensive data collection, unilateral policy modifications, automatic subscription renewals, broad liability disclaimers, or restrictive dispute resolution mechanisms. Such elements may impose tremendous risks to user privacy, monetary obligations, legal recourse, and service access continuity. Moreover, the discrepancy between contract performance and user awareness has resulted in a fundamental need for technological solutions to promote transparency and accessibility of legal aspects. Recent developments in Artificial Intelligence, mainly Large Language Models (LLMs) and Natural Language Processing (NLP), have shown enormous potential for automatic understanding of rich text. In addition, new methods, like Retrieval-Augmented Generation (RAG), semantic embeddings, and knowledge graph representations make structured understanding possible from unstructured documents (e.g., capturing semantic meaning and relational context). These advances are conducive to the building of intelligent systems, which are helping to support the comprehension of legal documents by users. The RiskWise framework, in this technological context, is presented here as an AI-driven solution to deal with legal documents, identify potentially risky clauses and present findings in a simplified and user-centric way. Using the hybrid form of retrieval and reasoning mechanism, the idea is to create a link between the complexity of a contract and user understanding to guide decision making in digital interactions.

1.2 Need Analysis

A lot of practical, technical and societal reasons for the need of an automated Terms and Conditions analysis system exist. First of all, contemporary digital ecosystems involve high levels of communication with a variety of online services, each one of which requires users to accept contractual agreements. Time-consuming and requiring legal literacy that non-standard users lack, reading these items by hand results in surface acceptance but not much consideration of the content. Second, entities often include clauses granting wide discretion such as the exclusive power to change clauses, the broad flexibility to share data and the rights to cancel accounts. When no analytical support is provided, users are exposed to unknowingly accepting unsatisfactory terms within a contract that may threaten their privacy, financial obligations to customers or their access to services. Thirdly, today’s automated applications are primarily focused upon providing a description of the work that users have done and fail to interpret the risk they have detected. Summarization may be an instrument to reduce the difficulty of documents but does not necessarily involve clauses introducing foreseeable potential risks (and why they might be associated). Consequently, users need systems which can not only minimize but also strategically identify, classify and situate contractual risks. Furthermore, the widespread global trend towards digital privacy and protection of consumers as well as compliance requirements demands that we find mechanisms that allow us to promote transparency to give the right of informed consent to participants. It allows the system to identify material clauses, categorize potential risks, and walks the user through an interactive exploration of contractual content, and greatly increases their knowledge and decision-making capacity. Thus, there is a great demand for the proposed intelligent, user-friendly and privacy-preserving system for the assistance of users to weigh legal contracts prior to acceptance.

1.3 Problem Identification

The most important problem addressed by the project is that users find it difficult to understand and assess complex legal documents in the digital service industry. This is due to a number of systemic issues. Firstly, legal documents contain specialized legal language and complex linguistic structures that are difficult to understand for laymen. Secondly, the nature of agreements is such that it deters complete reading, where selective attention (or complete avoidance) takes place. Thirdly, the average member of society does not possess the necessary legal knowledge to assess the contractual implications, such that it is unable to point out clauses that might have a potential impact on their rights and obligations. There are also no easy-to-use facilities to point out potentially unfair or risky clauses in contracts. The current state of affairs is such that it has low levels of interactivity, which might negatively impact the ability of users to pose questions or investigate specific contractual issues. All these issues mean that users lack awareness about agreements, which enables them to be vulnerable to privacy violations, financial liabilities, and service terminations with little understanding.

1.4 Problem Formulation

Building on the difficulties we have described here, this project explains the difficulty in terms of designing and putting into practice an intelligent system for automatically evaluating Terms and Conditions documents and giving users easily understandable, risk-aware insights. The suggested system must fulfil a number of usability and technical requirements in order to fully accomplish this goal. Working with documents in different formats, extracting useful semantic information from legal text, identifying and classifying clauses based on potential risk, and offering explainable reasoning to support risk assessments are some of the essential system requirements. To facilitate deeper user understanding, the system must also allow interactive querying and offer a trade-off between user privacy requirements and performance.

This problem was chosen because of its applicability, interdisciplinary characteristics, and possible implications for society. The objective of this project is to enhance user participation in digital decision-making processes and eliminate the complexity hurdle that comes with legal documents using modern NLP approaches, knowledge graphs, and interaction m1echanisms.

Chapter 2 - RESEARCH METHODOLOGY (OPTIONAL)

2.1 Research Approach

We use a hybrid research design combining the principles of design science research and applied research methodology in implementing this study. The main purpose of the work is to design, develop, and evaluate an intelligent program which analyzes Terms of Service (ToS) documents and helps detect potentially risky clauses. The research of design science is evidenced by an operative artifact, which can be known as RiskWise, that uses the latest techniques of Natural Language Processing (NLP), Knowledge Graphs, and Retrieval-Augmented Generation (RAG) for solving a real-world problem. Simultaneously, we see the elements of applied research, as the project aspires for a practical technological solution to the challenge of user comprehension of complex legal agreements. Research involved iterative problem understanding, literature exploration, system design, implementation, and validation through both experimental and case-based testing. This allowed for iterative improvements in system architecture and analysis pipeline based on insights obtained during development.

2.2 Preliminary Research and Literature Exploration

We did a preliminary literature review to get an idea of the current research on things like AI-assisted contract understanding, unfair clause detection, and legal document analysis. We acquired pertinent research articles and technical materials from academic databases and repositories (e.g., Google Scholar, arXiv, IEEE Xplore) via reference tracing from specific publications. The literature search focused on keywords such as Terms of Service clause detection, unfair ToS detection utilising NLP, legal document analysis, knowledge graph-based legal reasoning, Retrieval-Augmented Generation for legal documents, and Graph-RAG architectures. We looked at about 10 to 15 papers and chose 6 to 8 to read closely because they were new, relevant to legal document processing, methodologically innovative, and useful for explainable AI-driven legal analysis.

This exploratory phase helped us find important research areas that need more study, such as the fact that traditional NLP models are not good at structural reasoning, there are no clear ways to identify risks, and there is a need for hybrid search strategies that combine semantic similarity with relational reasoning.

2.3 System Development Methodology

The RiskWise system was made using a method that involved making prototypes and trying things out over and over. The project goes through a number of iterations. It starts with exploratory exercises using different clause detection models and ends with the design of a hybrid architecture that includes building a knowledge graph and RAG-based reasoning.

First, we did comparative analyses of transformer-based models for clause extraction tasks. This helped us understand the pros and cons of text-based methods. Based on these results, the system architecture was gradually improved to include structured ways to store legal information and ways to find it that take into account the context.

The iterative method made the system's parts modular, such as the document ingestion, semantic chunking, embedding generation, knowledge graph construction, hybrid retrieval logic, and generative reasoning modules. Every release of the work gradually improved the functionality of the features without changing how they worked with earlier versions. This way, the system's abilities can gradually improve.

2.4 Proposed Methodological Framework

The methodological framework of the project is based on a multi-stage processing pipeline designed to transform unstructured legal documents into an interpretable knowledge representation and risk analysis output. The major methodological stages are summarized below.

2.4.1 Document Ingestion and Semantic Parsing

It starts when users upload ToS documents in supported formats. Text extraction methods are used to extract raw textual content from documents, and semantic segmentation is performed into meaningful chunks. Semantic chunking, as opposed to fixed-length segmentation that discards contextual boundaries such as clauses and sections, can enhance the quality of downstream retrieval.

2.4.2 Embedding Generation and Vector Indexing

Semantic embeddings are obtained for each segment of the text chunk, based on our domain-adapted embedding models which can represent legal semantics. Similarity-based retrieval is performed by storing such embeddings in a vector index, which facilitates efficient similarity-based retrieval during query processing. The vector component of the system is responsible for fuzzy matching and semantic equivalence detection of legal terminology variations.

2.4.3 Knowledge Graph Construction

Alongside embedding generation, the system builds out a knowledge graph for the document. Large Language Models are used to derive subject-predicate-object triples that describe relationships between entities such as service providers, users, data, and obligations. These triples are mapped to a structured legal ontology and stored in a graph database, preserving relational and logical dependencies within the document.

2.4.4 Hybrid Retrieval Mechanism

In the system we implement a hybrid retrieval strategy involving vector similarity and knowledge graph traversal. Vector retrieval retrieves semantically relevant clauses, and graph traversal finds relationally connected information which don't share the common vocabulary. Incorporating these retrieval signals forms the comprehensive background information that further justifies the reasoning ability.

2.4.5 Generative Reasoning and Risk Analysis

A generative language model set up as a legal risk assessor gets the combined contextual information. The model analyses clauses, finds potentially risky parts of contracts, gives them severity scores, and makes natural language summaries that are easy to understand based on the context it finds. Using citations to prompt further increases the transparency and traceability of outputs.

2.5 Evaluation Strategy

The proposed system underwent a system evaluation utilising qualitative case-study analysis. We looked at a few Terms of Service documents with the developed pipeline to see how it works, how well it can find risks, and how well it can respond. Experimental phases culminating in comparative model evaluation have yielded additional empirical evidence for baseline clause extraction performance. A manual inspection of system outputs was conducted to confirm the relevance of identified clauses, the appropriateness of risk categorization, and contextual accuracy of generated explanations. Since the project was exploratory and prototype-based, it was thought that a qualitative validation method would be best.

2.6 Validation and Reliability Measures

To ensure the system's results are reliable and trustworthy, a method has been set up that includes several validation steps. Retrieval-Augmented Generation allowed the model to build its answers based on a document it had found instead of just using the knowledge it had already learned. Knowledge graph grounding made sure that clauses were consistent with each other, which made reasoning even harder. Other ways to check reliability included structured parsing of the information that was taken out, fallback mechanisms for dealing with model outputs that weren't consistent, and using supporting text segments in context. This combined method lowers the chance of hallucination and also makes it easier to understand the results of system-generated analysis.

Chapter 3 — LITERATURE SURVEY AND OBJECTIVES

3.1 Critical Review of Literature

A growing prevalence of digital services has resulted in a need for Terms of Service (ToS) agreements governing user-provider relations. But the complexity and lengthy nature of these documents have led to a number of significant research developments aimed at automated analysis of contractual material. The paper extends research on legal informatics, Natural Language Processing (NLP), and knowledge graph-based reasoning into interdisciplinary issues. Among these works, the CLAUDETTE framework is perhaps the most significant and introduced automated identification of potentially unfair clauses that form part of ToS agreements using ML techniques. The paper showcased that deep learning models can provide higher precision and recall over conventional classification methods. Although it came so far as a work of great significance to this space, the CLAUDETTE model was mostly based in clause-level classifications and did not handle contextual dependencies that could be captured between clauses, that is, does not reason on complex structures of contracts. Research along the lines of contract law has highlighted the broader societal consequences of unfair contract provisions and the need for regulatory oversight. As for unfair contract term legislation, research in the legal field of unfair contracts highlighted the gap between service providers and service consumers, highlighting the need for instruments that would facilitate transparent service provision and user awareness. Although these works offer a solid theoretical basis, they do not present automated technological tools for practical use by consumers. More recent work on the interface between AI and contract law has suggested neural and machine learning techniques for automated contract interpretation. These findings evidenced that AI-based systems can help to recognize unfair terms and assist lawyers with reviewing documents. Yet they also highlighted issues around interpretability, context-specific content, and embedding AI outputs within legal workflows. More recent breakthroughs using Retrieval-Augmented Generation (RAG) and graph-based reasoning have opened new applications to tackle these kinds of challenges. Structure-aware Graph RAG methods have brought to the fore the significance of representing legal documents as structured entities with hierarchical and relational relations to one another. These methods yielded an increase in contextual retrieval in contrast to vector-only methods, albeit they had an emphasis on legislative texts rather than private contractual agreements. Benchmarking studies measuring large language models for contract review also found deficiencies in long-context thinking and clause retrieval accuracy. Linguistic models may fail to recognise such information in large documents, leading to the interest in hybrid architectures that combine semantic-based retrieval with deterministic structural navigation. Hybrid Graph-RAG pipelines for applications including financial analysis have also been developed which integrate vector similarity retrieval with graph traversal to enhance multi-hop reasoning. Although encouraging, these methods were not adapted to the legal document analysis or for clause detection on the basis of risk. On the whole, the reviewed literature suggests a considerable improvement in automated contract analysis approaches while highlighting still outstanding structural reasoning, explainability, and user-centric risk interpretability issues. These insights directly guided the development of the RiskWise framework.

3.2 Literature Collection and Segregation

The studies evaluated in this study systematically identified academic repositories like Google Scholar, arXiv, IEEE Xplore, and reference tracing from relevant publications. The studies identified were categorized into thematic groups for analysis. In the first category, we identified studies on unfair clause detection and legal text classification, specifically the CLAUDETTE framework and other machine learning-related studies. These studies provided a basic understanding of risk identification using automation but were mostly confined to classification models. There were also studies related to legal and policy-related studies on unfair contract terms and consumer protection policies. These studies provided domain knowledge on various problematic clauses and their implications. The third category was studies on AI-based legal analytics and contract understanding systems, which discussed the development of neural language models and automated reasoning systems. The fourth category included studies related to Retrieval-Augmented Generation, Knowledge Graph integration, and hybrid retrieval models that have recently gained prominence. This study has been a methodological inspiration for the integration of semantic and structural models in document understanding. This thematic categorization allowed the identification of complementary studies among research areas, which also helped in identifying methodological gaps.

3.3 Summary of Literature

As per the literature review, automated legal document analysis is gradually shifting from traditional machine learning classification approaches to more complex language model-based approaches. Although designed to detect predefined patterns of clauses, classification-based approaches have a weakness in explanation and have tended to ignore clause interactions. Although it offers little computational help, the legal literature emphasizes the importance of dealing with unfair contractual clauses in society. AI-based approaches, on the other hand, tend to have contextual limitations and explanation difficulties despite their robust text understanding capabilities. In an attempt to improve context aggregation and reliability, new Graph-RAG models emphasize structural representations and multi-hop reasoning. They also propose hybrid retrieval approaches. However, there has been no extensive use of such models in consumer-friendly ToS analysis.

Thus, in conclusion, the literature combined suggests the viability of automated ToS analysis and the need for systems that integrate structural reasoning along with semantic retrieval and explainable risk analysis.

3.4 Identification of Gaps

Based on the critical review and synthesis of the literature surveyed, some research gaps were identified that form the basis for the development of the proposed system.

First, current systems for unfair clause detection are largely dependent on classification models that examine clauses independently. Although these models are successful in identifying clauses belonging to predefined categories, they are often not able to take into consideration the larger context of the document in which the clauses are presented. As a result, the relationships between clauses, supporting statements, and cross-references that may have an impact on interpretation are often not considered, making it difficult to arrive at a reliable risk assessment.

Second, current AI-powered contract analysis tools are often not very transparent about their results. Although these tools are successful in identifying clauses that may be potentially problematic, they are often not very forthcoming about the rationale used for risk classification.

Thirdly, traditional Retrieval-Augmented Generation tactics typically use a flat text file to represent a document. Because of this, close and related semantic meanings of phrases and clauses are likely to become dis-sociated or fragmented. Because of this fragmentation of relevant context, critical and related information will not be considered in reasoning, potentially leading to incomplete reasoning when using generated insights.

Fourth, current research using large large-scale Language Models indicates an inability to correctly attribute key-elements or concepts found in long-form documents, where complex, interconnected, and hierarchical structures exist. When interpreting or analyzing interrelated clauses in an agreement, an LM may find it challenging to maintain the same level understanding and sequential context through the entire document, which can lead to incomplete retrieval and/or partially attributed understanding of the relationships between the contracts.

Finally, most of the available research currently devoted to automated contract analysis Technologies focus on tools designed specifically for legal practitioners or large-scale enterprise business users. The technological capabilities offered by the majority of these tools can only effectively be utilized by users who have the relevant subject-matter expertise, as a user may receive output in either a technical format or a legal format, that is not an intuitive or user-friendly means of delivery of actionable and interpretive conclusions.

3.5 Need for the Project

The gaps identified above indicate that there is a noticeable need for a smart system, providing semantic and structural representations on ToS documents and explaining results to non-expert users. Such a system would be beneficial for increasing transparency in digital agreements, promoting informed consent, and increasing consumer awareness.

Additionally, the increasing usage trend of generative AI can be leveraged to develop hybrid architectures targeting limitations of current techniques. This project seeks to bridge the gap between automated detection and human-understandable explanations of contractual risks by integrating knowledge graph reasoning with RAG-based language models.

3.6 Problem Statement

Users of digital services are unaware or unprepared to decipher the extensive amount of complex contract language present within terms (terms) of service agreements existing on the digital platform. The majority of users also do not think about the contractual terms before they accept the terms. There is a large discrepancy between user acceptance of terms of service agreements and user awareness of the terms of service agreements. This is one of the primary threats to the user making informed decisions about services provided by digital platforms as well as limiting transparency for users within an entire digital ecosystem. While a few automated solutions have been explicitly directed toward performing contract analysis of digital service contract terms, many automated solutions currently in existence have significant flaws. Many of the automated solutions relying on a clause-based classification system to analyze service contract terms for digital platforms cannot accurately identify or provide meaningful context for the resolution of discrete clauses and the resolution of multiple clauses that exist within an individual service contract. Certain automated solutions use automated summarization models based on language modeling that rely on non-specific models to identify and provide context for discrete clauses associated with producing a potentially risky contractual obligation. Consequently, there is a need for an automated solution that is capable of mediating service terms agreements using contextual and structured problem resolution methods and user-centered interpretation. Because of this need, this project will submit a proposal for continuing research and development of an automated solution for the purpose of assisting users in reading and understanding their service terms contracts on the digital platform.

3.7 Objectives

The main aims and objectives of the project are as follows:

1. Design and develop a hybrid retrieval system that combines vector similarity search and knowledge graph traversal for legal document analysis.

2. Design and develop an automated system that can extract structured knowledge representations from unstructured Terms of Service documents.

3. Design and develop a system that can identify and classify potentially risky contractual clauses using context-aware reasoning.

4. Design and develop a system that can provide explainable natural language interpretations of identified risks.

5. Design and develop an interactive web-based interface that can allow users to upload documents and perform automated risk analysis.

3.8 Limitations of the Study

The study has a number of limitations. It is based on language models, whose performance may depend on the capability of the model and the complexity of the documents. The system is mainly suited for English-language documents and is not applicable in multilingual settings. In addition, the study has used qualitative methods of evaluation because the study is exploratory in nature, and large-scale quantitative benchmarking is not within the scope of the study.

Resource constraints related to local model execution and the variability of LLM outputs are also limitations of the current system.

Chapter 4 — ACTUAL WORK

4.1 Methodology for the Study

By implementing the RiskWise system based on a multi-stage process, an unstructured legal document like a Terms of Service is converted to a structured knowledge representation, which is accompanied by user-related risk insights. Unlike the research methodology outlined in Chapter 2, which highlights more on conceptual design and research strategy, this section describes the actual operational workflow developed in the developed framework. The processing pipeline is intended to transform raw legal text into understandable outputs in a systematic way through a series of structured analytical steps.

The method consists of document ingestion where users upload Terms of Service documents via the web-based interaction. Backend service is used to evaluate uploaded files and to handle documents in multiple formats, ensuring flexibility and usability across diverse sources. After ingestion the text extraction process extracts the textual data from the documents and maintain logical order and structural integrity. Next, we perform semantic segmentation of the extracted content (coherent chunking of text at clause level). This segmentation mechanism maintains ongoing context and facilitates the efficient retrieval and reasoning tasks performed further down the line.

Next, semantic embeddings are created for each segmented text unit using domain-adapted embedding models appropriate for representing subtle legal semantics. These embeddings are vectors of textual meaning allowing similarity-oriented retrieval over semantically related clauses even beyond lexical overlap. The generated embeddings are stored in a vector indexing mechanism that allows fast and scalable nearest-neighbor search during analysis and query processing.

Alongside embedding generation, language models can also be used to extract subject-predicate-object triples from textual segments, which in turn form the groundwork for a representation in a knowledge graph. Such a structured representation can account for relational dependencies among the entities like users, service providers, permissions, obligations and contractual actions. By establishing relationships between document components, the knowledge graph makes relational analysis and context connection possible beyond surface patterns of textual similarity, enhancing the semantic knowledge of contractual content.

In addition, to combine semantic knowledge with that of the structure, a hybrid retrieval mechanism is designed to use vector similarity search together with knowledge graph traversal. This integrated method allows one to retrieve contextually relevant clauses together with relationally connected information, thereby yielding enriched contextual assemblies that allow comprehensive analysis. The hybrid retrieval strategy combats the shortcomings of single-modality retrieval methods while also improving the richness of contextual evidence provided for reasoning tasks.

Lastly, the contextual information retrieved is processed by a generative reasoning module set up for risk-based processing. This module provides analyses of the context, and uses it to categorize potential risky clauses, to make severity judgments to help derive a score and categorize risk-types and to provide explainable summaries in natural language based on evidence retrieved. Combining retrieval grounding with generative reasoning, the system's outputs are interpretable, contextually aware, and consistent with document details.

The RiskWise system manages the efficient, explorable and context-sensitive analysis of contractual documents through the multi-step and coordinated approach, making it possible to process intricate legal agreements into general users’ accessible results.

4.2 Experimental Work

The experimental stage of the project was characterized by iterative evaluation of different methods for automated clause detection and risk identification. Initial experimentation explored transformer-based models to perform question-answering tasks for clause extraction, giving baseline insights about what the models can do for understanding legal text. Subsequent experimental phases focused on validation of prominent components across different architectures, such as semantic chunking schemes, accuracy of embedding-based retrieval and knowledge graph construction. ToS documents from a variety of digital platforms were examined and processed in the pipeline to analyze system robustness for different document structures and legal formulations using multiple ToS documents as case studies. Overall, the comparative analysis of the retrieval results confirms that hybrid retrieval mechanisms achieved greater contextual completeness than purely semantic retrieval. Also, testing local and cloud-based inference modes allowed for the comparison of performance trade-offs between computational efficiency and response latency. The experiments informed architectural fine-tuning and supported validation of the proposed hybrid Graph-RAG framework.

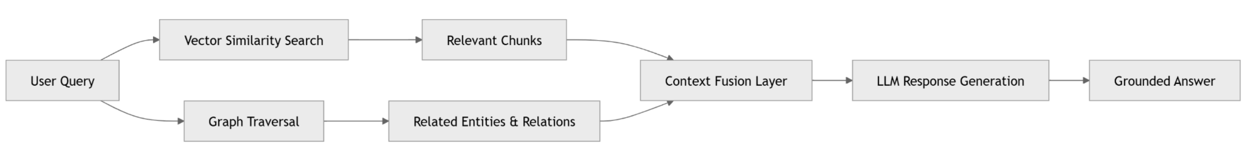
4.3 Analytical Work

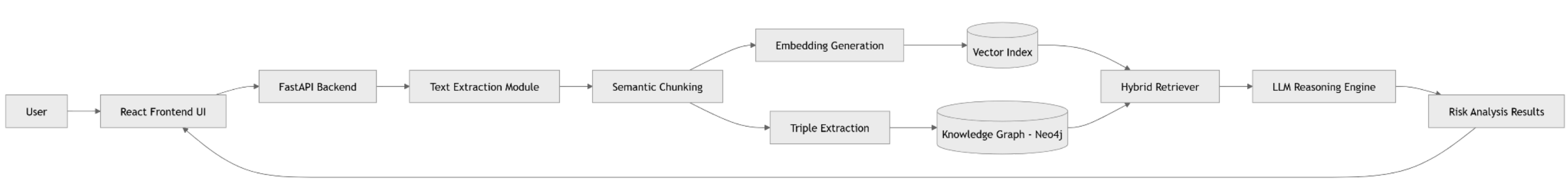
Analytical work performed in the project mostly focused on analysing system outputs to address relevance, contextual integrity, and interpretability of the generated risk insights. We manually reviewed the retrieved clauses for semantic similarity search for contextually appropriate segments corresponding to user queries and analysis tasks. The extracted relationships were checked to be accurate in terms of logical consistency to ensure that the extracted relationships corresponded to that of identified entities. Graph traversal results were checked to confirm that relational context contributed meaningful information beyond vector retrieval results. The risk analysis outputs created by the reasoning module were checked for clarity, correctness, and alignment with the underlying document content. Severity scores and risk categorizations were observed to see if these actually represented matters of contractual concern, such as unilateral modifications, limitations on liability, and data usage permissions. From this analysis, the project ensured proper contextual grounding and user interpretability of system outputs.

4.4 Modeling, Analysis and Design

A modular and multi-tiered architecture is used to represent the system. The system consists of an end-user web-based interface (presentation layer), a back-end processing system (processing layer), and storage system (storage layer). The focus of the presentation layer is to provide an intuitive and simple-to-use interface for users (non-technical). The processing layer is where the intelligence of this system resides and consists of six processing modules: text extraction, semantic chunking, embedding generation, knowledge graph construction, hybrid retrieval, and generative reasoning. While each module functions independently, they all contribute to the same processing pipeline. The storage layer utilizes a graph database that has vector indexing and relational storage capabilities.

Furthermore, this design aids in retrieval and reasoning by attaching semantic and structural information together. The key factors in how the system was designed are modularity, scalability and clarity. Because the system separates the vector and graph representation of information, different retrieval options can be utilized. The modular processing components can accommodate future growth and enhancements to the processing models without the need to re-design the entire system.





4.5 Prototype and Testing

To demonstrate end-to-end analysis functionality, the RiskWise system has been created as a prototype consisting of a web user interface integrated with back-end capabilities to process and analyze documents. Users can upload ToS documents, check their processing status, view risk results classified by type, and interact with the system using a conversational query interface. The prototype testing was performed by simulating various user scenarios (e.g. uploading a document, displaying risk analysis results, and answering queries in context). Functional testing was performed to verify that each stage of the pipeline was executed correctly and integration testing was conducted to verify that the front-end and back-end systems interacted correctly. For performance testing, processing time was measured for different document sizes and inference configurations to ensure there were expected differences in latency between local and cloud executions. Unsupported formats and invalid inputs were included in the testing to validate error handling and confirm strength of the system. The overall prototype testing indicated compliance with the functional correctness and usability requirements of the system.

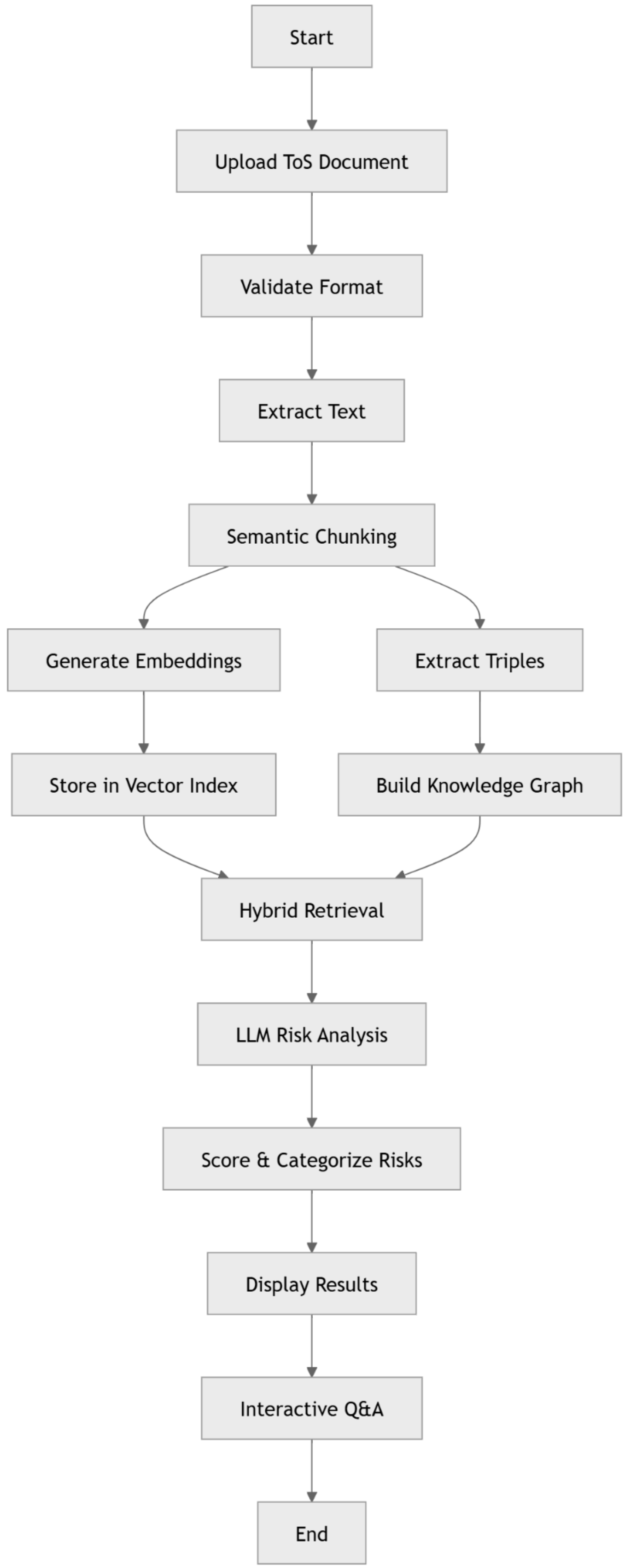
4.6 Algorithms and Flowcharts

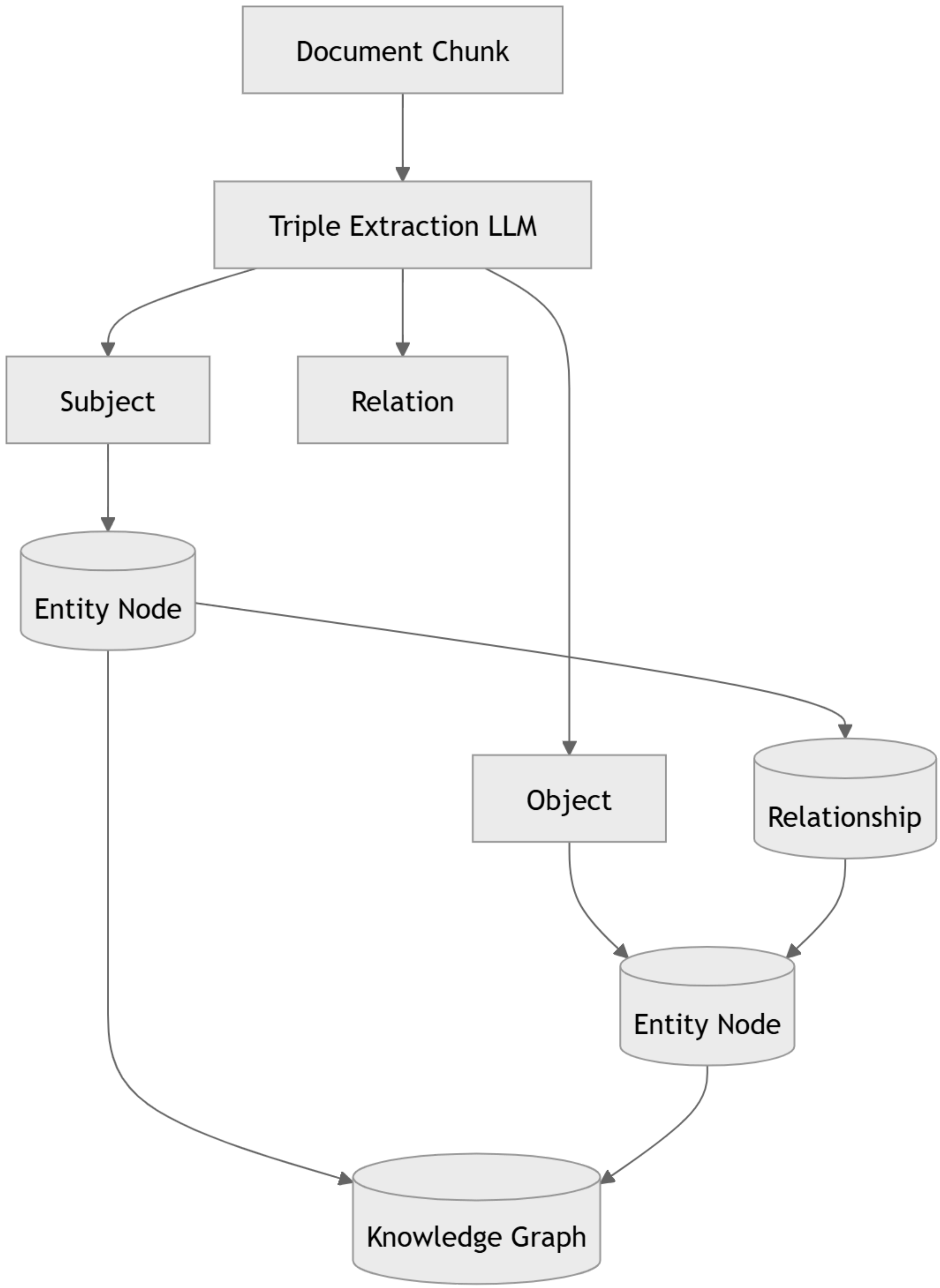
The operational workflow of the RiskWise system can be represented through algorithmic steps describing the document analysis pipeline.

Algorithm: RiskWise Document Analysis Pipeline

1. Start
2. Accept document upload from user
3. Validate file format and generate document identifier
4. Extract textual content from document
5. Perform semantic segmentation into text chunks
6. Generate embeddings for each chunk
7. Store embeddings within vector index
8. Extract knowledge triples from chunks

9. Construct knowledge graph representation
10. Perform hybrid retrieval combining vector similarity and graph traversal
11. Provide retrieved context to generative reasoning module
12. Identify risky clauses and assign severity scores
13. Generate explainable summaries and present results
14. Enable interactive user querying
15. End


Start
Upload ToS Document
Validate Format
Extract Text
Semantic Chunking
Generate Embeddings
Extract Triples
Store in Vector Index
Build Knowledge Graph
Hybrid Retrieval
LLM Risk Analysis
Score & Categorize Risks
Display Results
Interactive Q&A
End


Document Chunk
Triple Extraction LLM
Subject
Relation
Object
Entity Node
Relationship
Entity Node
Knowledge Graph

4.7 Codes and Standards

RiskWise system was developed in accordance with recognized standards of software engineering, with established tools in data handling, industry-oriented practices and standard methods to maintain reliability, integrity, maintainability, compatibility and responsible behavior. Due to being a work in the automated legal report inspection area, transparency and modularity and explainability at all stages of development were the focus of the code.

Standards for Software Development

The framework was mostly developed in Python code and was written in a standard coding fashion to make it easier to read, collaborate, and maintain. To make sure that the code was readable, standard naming and modular programming practices, as well as organizing modules in an organized way, were followed. The main functional modules, such as document processing, embedding generation, knowledge graph construction, retrieval logic, and reasoning modules, were developed as separate modules.

API and Integration Standards

Both front-end-side and backend layer functionality of the interaction between pieces of system programming was handled by creating code on, for example, RESTful API principles of the RESTful API framework to enable communication between API components. Data exchange was done through standard HTTP request-response mechanisms, and JSON format which the data was used as the main data format because of its lightweight layout and the ability to use this media type for its lightweight and applicable platform. These integration methods were leveraged to facilitate easy integration of document-based service-ingestion systems, retrieval pipelines, retrieval methods, and interfaces in the user interface components were optimized by seamless orchestration.

Data Management and Data Protection Standards

Since user uploaded documents could include potentially proprietary contractual information, privacy-aware data handling practices were accounted for in the system development. We used uploaded documents as workstopped files and did not keep superfluous metadata and used temporary caching mechanisms for each iteration of the same document to enhance performance. The approach is focused on minimizing data and treating analysis artifacts with control; both data protection principles and ethical AI development principles are well-accepted.

Ethics in AI and Responsible Usage Guidelines

The RiskWise system was developed with a focus on responsible AI use. The returns on risk analysis are structured in a way that is based on the document context retrieved to mitigate the issue of hallucination in terms of how the generated explanations can be made more transparent in terms of interpretation. The system has a purposeful approach to its use in promoting user awareness rather than legal advice, and the generated outputs are provided in the form of informative insights that are meant to aid understanding.

NLP Standards and Knowledge Representation Standards

Normalization methods in NLP, such as text normalization, sentence segmentation, and semantic chunking, are performed on a periodic basis for data preparation in NLP. The embedding methods used were based on the transformer modeling paradigm that has been popular in modern semantic retrieval models. Additionally, the development of the knowledge graph was based on the subject-predicate-object representation paradigm that has been widely used in knowledge engineering, enabling data structure relationships in all document materials.

System Design and Documentation Standards

Project documentation is guided by academic and engineering documentation conventions for a modular system description, architectural diagrams, algorithmic representation, experimental verification and validation. We included flowcharts and system diagram of system functionality, which will facilitate reproducible and technically transparent design, so that the system design can be understood, reviewed and extended by any future researcher or developer.

Following these coding, integration and operational conventions, RiskWise system ensures enhanced reliability, scalability and interpretability while ensuring the responsible processing of user provided contractual data. These behaviors all facilitate the development of a scalable and resilient platform for in the automatic interpretation of legal documents.

Chapter 5 — RESULTS, DISCUSSIONS AND CONCLUSIONS

5.1 Results and Analysis

A series of case-based experiments based on Terms of Service documents gathered from various online platforms were used to assess the RiskWise system. Using its hybrid pipeline, the system processed the uploaded documents to create knowledge graphs, produce semantic representations, and provide risk-informed insights based on the contextual evidence gathered. Based on observations from the system implementation and experimentation as well as the synthesis of the literature, the results are discussed.

5.1.1 Findings of Literature Survey

The literature survey conducted as a part of this study has been very informative in understanding the status of automated contract analysis and research in legal AI. The following points have been raised based on the studies surveyed:

• The current state of unfair clause detection is based on classification models analyzing clauses independently.

• Most AI-based contract analysis tools lack contextual models that understand the relationships between clauses.

• Lack of explainability is a problem, and most tools are able to detect risks but lack the ability to explain their reasoning.

• The traditional Retrieval-Augmented Generation model analyzes text in a flat structure, which can lead to the loss of contextual information.

• Most tools are designed for legal professionals and not end-users.

Analysis:

The literature survey conducted emphasizes the need for a system that bridges the gap between automated clause detection and user-friendly explainable analysis. The RiskWise framework fills this gap by incorporating hybrid retrieval based on semantic similarity and relational reasoning.

5.1.2 Results of System Implementation

5.1.2.1 Contextual Retrieval Performance

• The segmentation of the uploaded documents was done into semantically significant chunks.

• Vector space representation made it easier to retrieve similar clauses based on similarity.

• Traversal of the knowledge graph helped to expose contractually relevant information.

• Hybrid retrieval combined the results to create contextual evidence.

Result: Improved contextual completeness and efficient retrieval of distributed contractual provisions.

5.1.2.2 Risk Detection and Explanation Quality

The ToS risk identification system was tested using example ToS documents to determine the functionality of the system in identifying risks associated with typical ToS documents.

• Unilateral modification clause identification

• Liability limitation clause identification

• Data usage and privacy permissions identification

• Termination and service restriction clause extraction

The explanations provided were tested for their relevance and interpretability.

Result: Risky clauses were successfully extracted along with relevant explanations.

5.1.2.3 Interactive Querying Performance

The conversational interface was tested to determine the capabilities of user interaction and exploration of context.

• Users were able to ask follow-up questions about risks identified.

• Contextual evidence retrieved supported informed response.

• Responses were still relevant to document content despite turns in interaction.

Result: Successful exploration of contractual risks using the conversational interface while still grounded in context.

5.1.3 Overall Observation

The combined results demonstrate that the RiskWise system successfully transforms complex legal agreements into structured representations and interpretable insights. The hybrid Graph-RAG pipeline enabled context-aware retrieval and explainable reasoning, validating the system’s suitability for assisting users in understanding contractual risks associated with digital services.

5.2 Comparative Study

This study compares a traditional approach for clause extraction, text only; and a newly proposed hybrid Graph-Recurrent Attention to Graph (RAG) Model for clause extraction. The existing literature demonstrates that transformer based question-answering models exhibit reasonable efficiency for clause extraction but fall short of contextual completeness and relationship oriented reasoning. Conversely, the proposed hybrid model in the RiskWise framework performed better with respect to capitalising upon contextual context, and demonstrating relationship between clauses. In addition, the knowledge graph aspect of the model facilitated identification of obligations/permissions which related but were not derived from semantic similarity. Also, the hybrid model provides contextually appropriate citation for both the text segments derived from clauses and the evidence for the relationships, making them contextually interpretable as well. Ultimately, this comparative analysis demonstrates that combining structural representations of contracts with semantic extraction improves reliability and interpretability in automated contract analysis systems.

5.3 Discussions

The proposed project clearly demonstrates the applicability of the knowledge graph and Retrieval-Augmented Generation paradigm to be combined for addressing legal document understanding tasks. By combining semantic as well as relational approaches, the system bridges the gap that has been noticed in the past. It is also important to note that the structural representation helps in improving the contextual completeness, especially when related clauses are identified in different spots. Additionally, the conversational interaction framework helps in improving accessibility by allowing users to inquire about document understanding in natural language queries. On the other hand, the study also points out some intrinsic limitations of generative AI systems, such as variations in output and reliance on model capability. Other trade-offs in performance between local and cloud inference modes have also been noticed, which are indicative of real-world deployment aspects.

5.4 Project Cost Estimation

The RiskWise project was primarily developed using open-source technologies and existing computational resources, resulting in minimal direct financial expenditure. However, a conceptual cost estimation is presented to reflect potential implementation and operational expenses.

Project Execution Cost

No	Category	Description	Estimated Cost (₹)
1	Software Tools	Open-source frameworks and libraries	0
2	Hardware Resources	Personal computing resources	0
3	Cloud Usage	API inference and hosting (estimated)	2000
4	Miscellaneous	Testing and internet usage	1000
Total			**3000**

Service Deployment Cost (Hypothetical)

No	Component	Estimated Monthly Cost (₹)
1	Cloud hosting	1500
2	Model inference API	3000
3	Database hosting	1000
Total		**5500**

5.5 Project Impact

5.5.1 Environmental Impact

This application is based mainly on software and does not involve large amounts of physical resources. Cloud-based processing may increase indirect energy consumption. However, due to clever retrieval mechanisms and caching strategies for the best possible retrieval, there is less effort for computer overhead because redundant computation is avoided.

5.5.2 Societal Impact

The significance of the RiskWise project as a whole is on the side of humanity. The system encourages transparency and informed consent in digital environments. The system assists users to make educated choices regarding digital service adoption through comprehension of terms, by reducing the complexity of contractual contracts. It helps make a dent in consumer knowledge, enhances digital literacy, and gives everyone equitable access to information.

5.5.3 Economical Impact

Economically, the system demonstrates potential to reduce legal consultation costs for preliminary contract understanding and risk identification. Organizations may also benefit from automated contract review capabilities, improving operational efficiency. The architecture's reliance on open-source technologies further enhances cost-effectiveness and accessibility.

5.6 SDG Goal Compliance

The project satisfies a number of the United Nations Sustainable Development Goals (SDGs). The project satisfies SDG 9 (Industry, Innovation, and Infrastructure) through the development of innovative AI-driven infrastructure for automating document analysis. The project also satisfies SDG 16 (Peace, Justice, and Strong Institutions) through the promotion of transparency and fairness in contractual relationships and through the enhancement of access to understandable legal information. The project indirectly supports the promotion of inclusive digital ecosystems and responsible technological innovation through the empowerment of users with knowledge about digital agreements.

5.7 Conclusions

The current study has been able to achieve the objectives of designing and implementing a hybrid Graph-RAG approach for the automated analysis of Terms of Service. The system has been able to efficiently combine semantic embeddings, knowledge graphs, and generative reasoning power.

The RiskWise system designed in the current research study demonstrates the feasibility of transforming complex legal texts into interpretable insights using context-aware analysis and reasoning. The hybrid search strategy was able to enhance the completeness and correctness of context awareness over the baseline approaches, and the conversational interface was able to enhance accessibility for non-experts.

The current research study introduces a new contribution to the legal AI community by offering a new consumer-oriented application of hybrid search architectures.

5.8 Scope for Future Work

RiskWise sets up a base framework for automated Terms of Service analysis based on hybrid retrieval and reasoning methodologies. Nevertheless, there are still many research and development opportunities to improve system capability and expand its flexibility as well. Immediate direction is the integration of multilingual processing to support analysis of contractual documents even in several languages. The use of multilingual embedding models and language-agnostic large language models will ensure that the user base will be global in nature, and also expand the usability beyond English-language agreements.

In addition, the incorporation of Optical Character Recognition (OCR) processing would also allow scanned and image-based documents to be processed and thus extending the coverage on document-based applications and its real life relevance.

Another major extension is the ability to visualise knowledge graph structures more graphically. Interactive and dynamic graph visualizations would enable users to explore the relationship and the implications of contractual entity vs obligations vs permissions that makes them more interpretable and can improve understanding in terms of how to understand the document semantics. This visualization might additionally benefit legal practitioners in exploratory contract analysis and compliance auditing.

Personalization methods may also be investigated under the Risk Analysis model in future. Customizable risk profiles would make it so users can prioritize individual categories - privacy, financial obligations, liability clauses, and so on. Adaptive analysis that caters to user preferences may help in enhancing relevance and providing context aware interpretation of contractual risks.

Integrating with browser extensions is a promising line of real-time ToS evaluation from a deployment perspective. This kind of integration could automatically analyze agreements encountered while interacting online, offering proactive intelligence before one agrees a contract's language terms. Also, enterprise-oriented batch processing capabilities could let companies review multiple contracts with various vendors and perform comparative contract risk assessments.

Methodologically there are opportunities to broaden the reasoning ability of the enterprise, by including legal terminology and databases for their domain knowledge. For example, augmenting the construction of knowledge graphs with external legal standards and precedent information could allow semantic richness and enable more complex compliance based reasoning.

The research agenda's focus will also be to engage in empirical validation of the proposed framework by conducting quantitative benchmarking with annotated legal datasets and evaluating system performance metrics related to clause identification and retrieval accuracy: identifying user studies to support research into usability, trust and interpretability can also support determining user performance in human-centered systems. The ability to consider and investigate possible distributed infrastructures for data processing, optimally indexed storage technologies, and compression techniques will facilitate broad-scale application opportunities. Enhancements to privacy and security through on-device inference and encrypted knowledge storage will also create greater user trust and increase user experience. RiskWise is a scalable platform that will continue to enable more research into AI-enhanced contract comprehension. These types of research initiatives are decentralizing the direction of hybrid retrieval architectures research and pursuing paths that will produce real-world deployment environments that have embraced transparency and will yield results of enlightenment.

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Publication Details

No publications have arisen from this project work at the time of report submission. The work presented in this report represents original research conducted as part of the Bachelor of Technology degree requirements.

Appendix B — Hardware and Software Tools

B.1 Hardware Requirements

The RiskWise system was developed and tested using standard personal computing infrastructure capable of supporting modern web development and machine learning workloads. The hardware configuration used during development is summarized below.

Component	Specification
Processor	Multi-core CPU (Intel i5/i7 or equivalent)
RAM	16 GB system memory
Storage	SSD-based storage for faster indexing and retrieval
GPU (optional)	Not mandatory; used only if local LLM inference required
Network	Stable internet connection for cloud API access

For privacy-focused deployments involving local model execution, higher RAM capacity and GPU acceleration may be beneficial to ensure acceptable inference latency.

B.2 Software Requirements

The RiskWise system employs a full-stack architecture integrating multiple software components across frontend, backend, data storage, and AI processing layers.

Category	Software / Tool	Purpose
Operating System	Windows / Linux	Development and execution environment
Programming Language	Python	Backend and AI pipeline implementation

Frontend Framework	ReactJS	User interface development
Backend Framework	FastAPI	API and asynchronous processing
Database	Neo4j	Knowledge graph and vector storage
AI Framework	LangChain	Retrieval orchestration and LLM integration
NLP Libraries	spaCy, Transformers	Text processing and embeddings
Version Control	Git & GitHub	Source code management
Package Manager	pip / npm	Dependency management
Development Environment	VS Code	Code editing and debugging

B.3 Deployment and Runtime Tools

Additional tools supporting system execution and experimentation include:

Tool	Purpose
Ollama	Local LLM inference
Groq API	Cloud-based LLM inference
Node.js	Frontend build environment
Browser	User interface access and testing
Mermaid / Diagram tools	System visualization

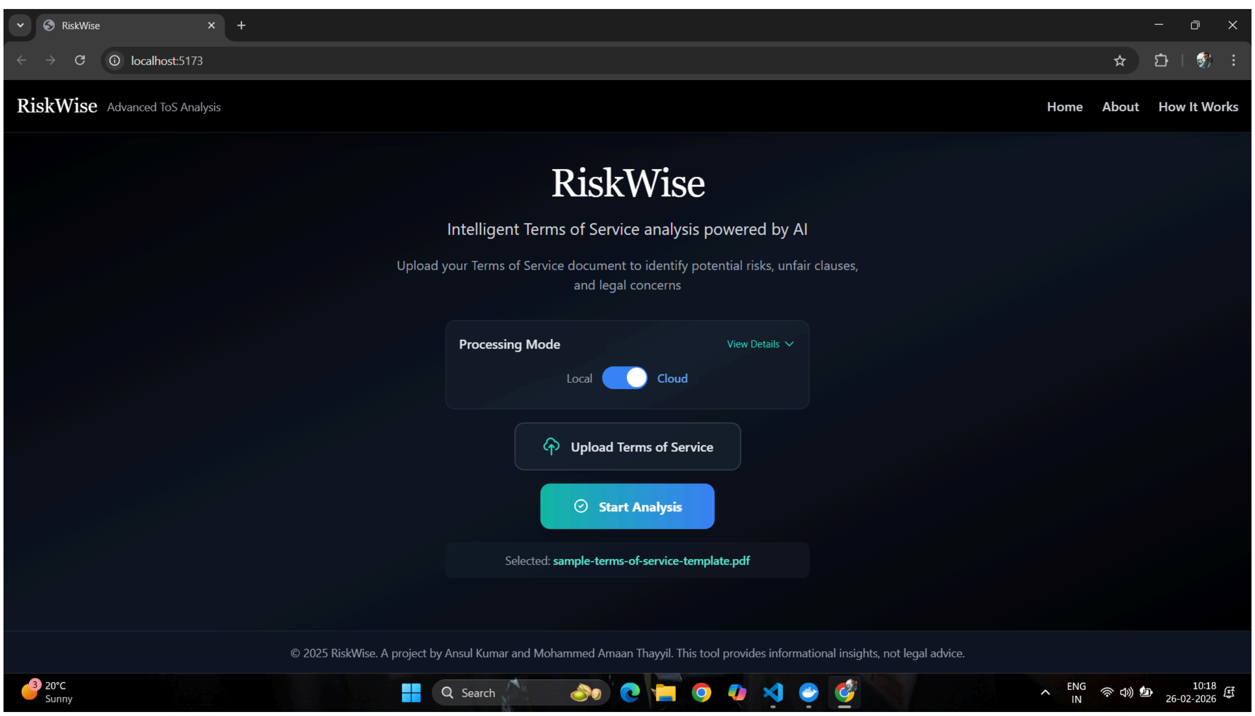
Appendix A — Supporting Material

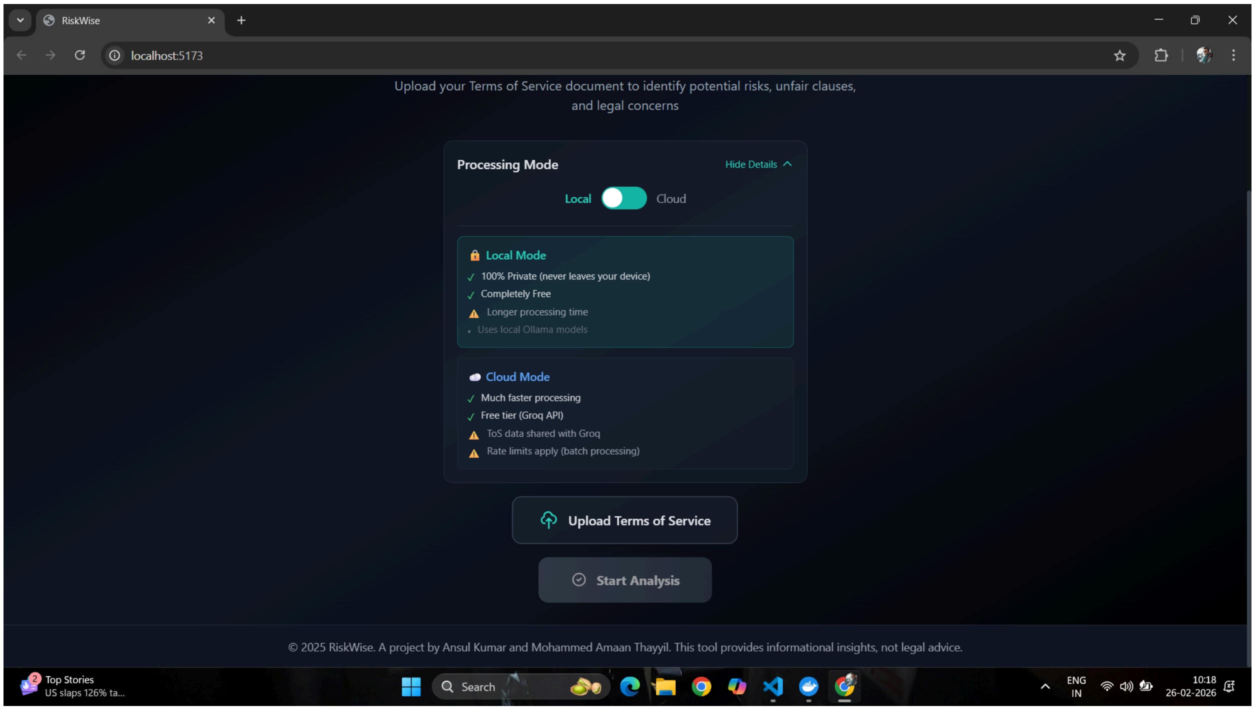
A.1 Sample User Interface Screens

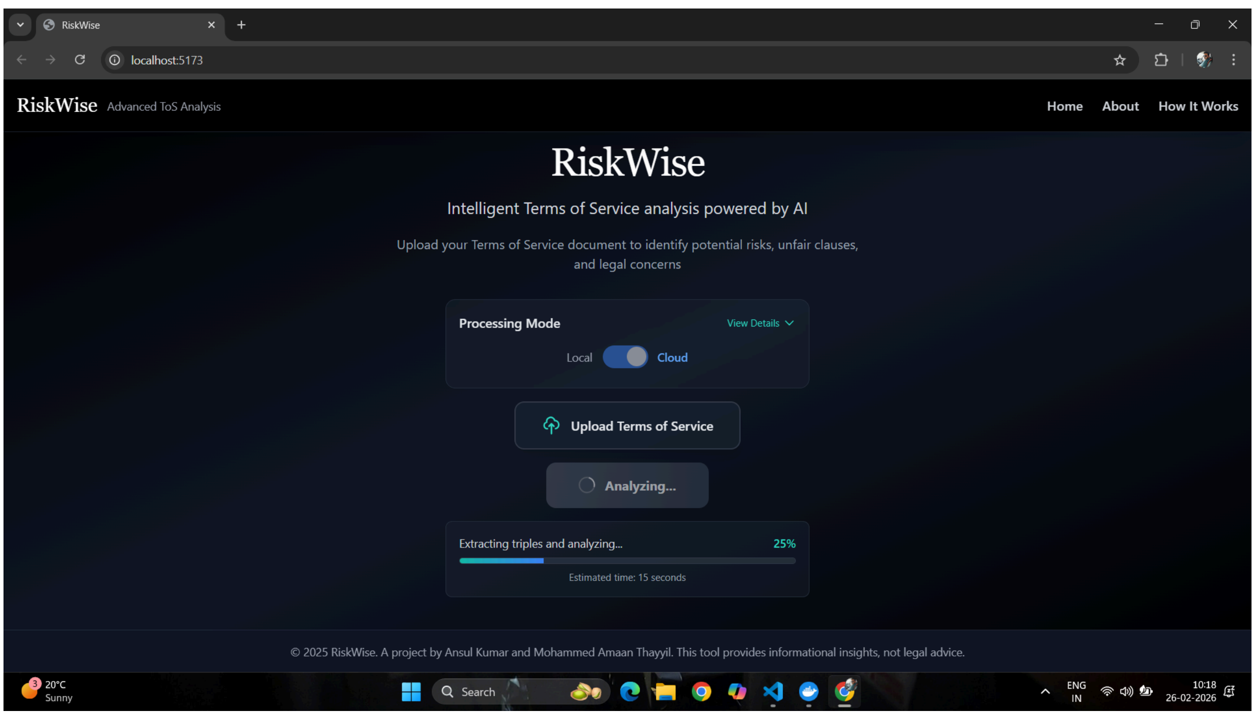
The RiskWise system provides a web-based interface that enables users to upload Terms of Service documents, view analysis results, and interact with the system through a conversational interface. Representative screenshots illustrating major system functionalities are included below.

A.1.1 Document Upload Interface

This screen allows users to select and upload Terms of Service documents for analysis. The interface includes options to choose processing mode and initiate analysis.

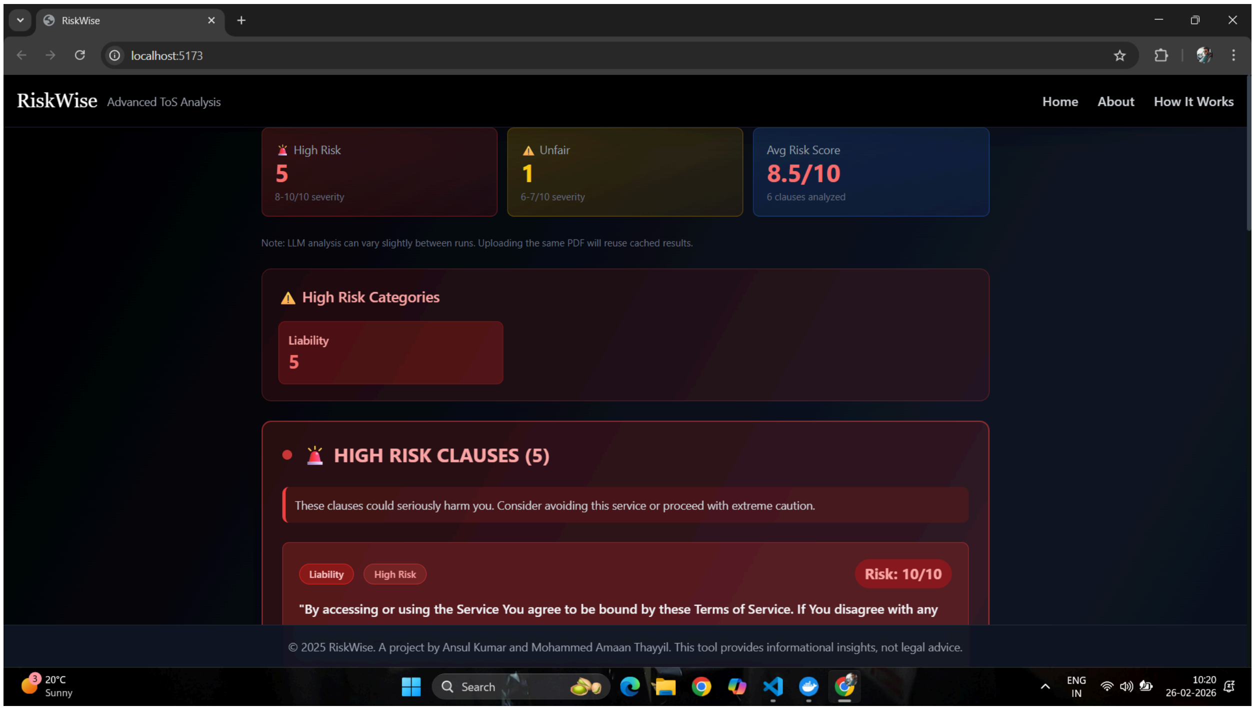


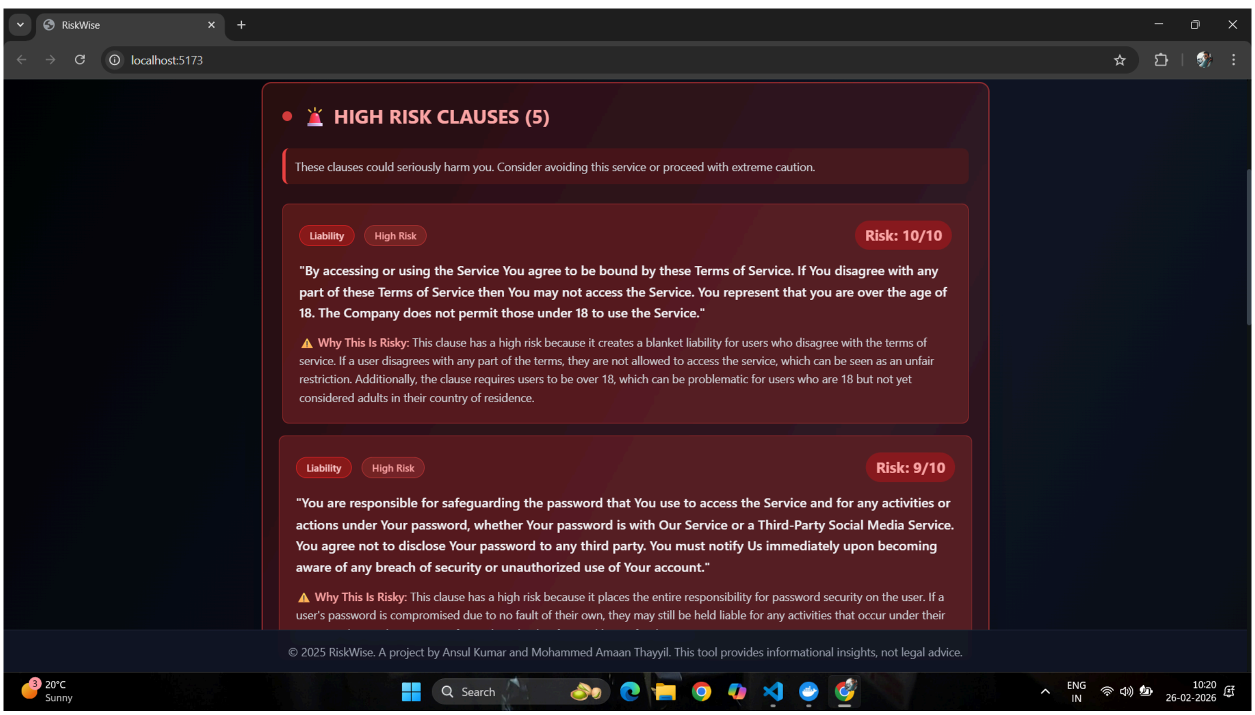


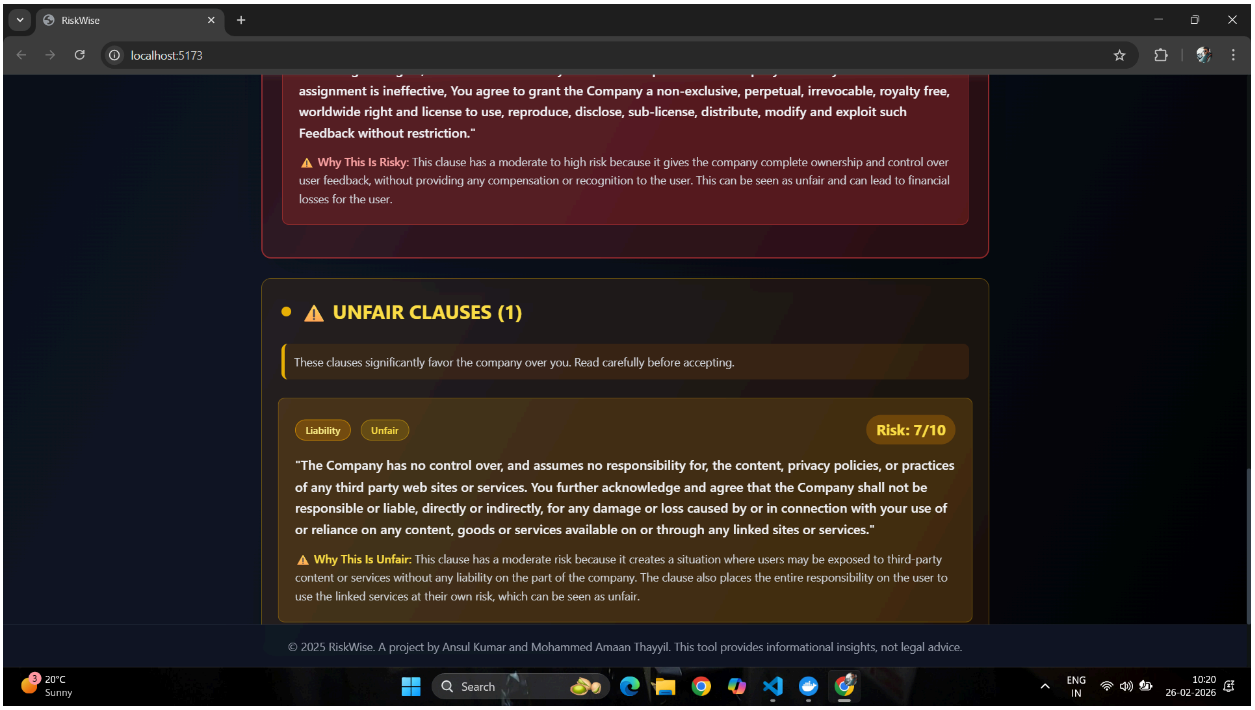


A.1.2 Risk Analysis Results Interface

Following document processing, the system displays categorized risk insights highlighting clauses of potential concern along with severity scores and explanations.

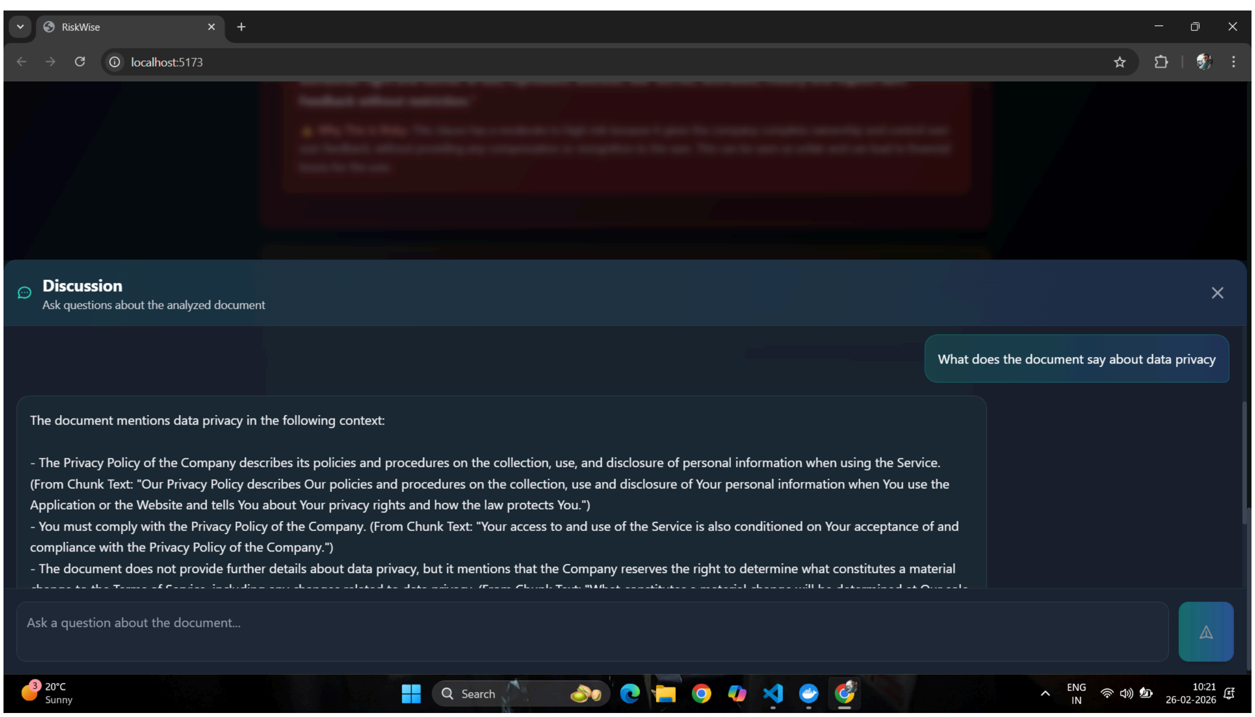

RiskWise
localhost:5173
RiskWise Advanced ToS Analysis
Home About How It Works
High Risk
5
8-10/10 severity
Unfair
1
6-7/10 severity
Avg Risk Score
8.5/10
6 clauses analyzed
Note: LLM analysis can vary slightly between runs. Uploading the same PDF will reuse cached results.
High Risk Categories
Liability
5
HIGH RISK CLAUSES (5)
These clauses could seriously harm you. Consider avoiding this service or proceed with extreme caution.
Liability High Risk
Risk: 10/10
"By accessing or using the Service You agree to be bound by these Terms of Service. If You disagree with any
© 2025 RiskWise. A project by Ansul Kumar and Mohammed Amaan Thayyil. This tool provides informational insights, not legal advice.

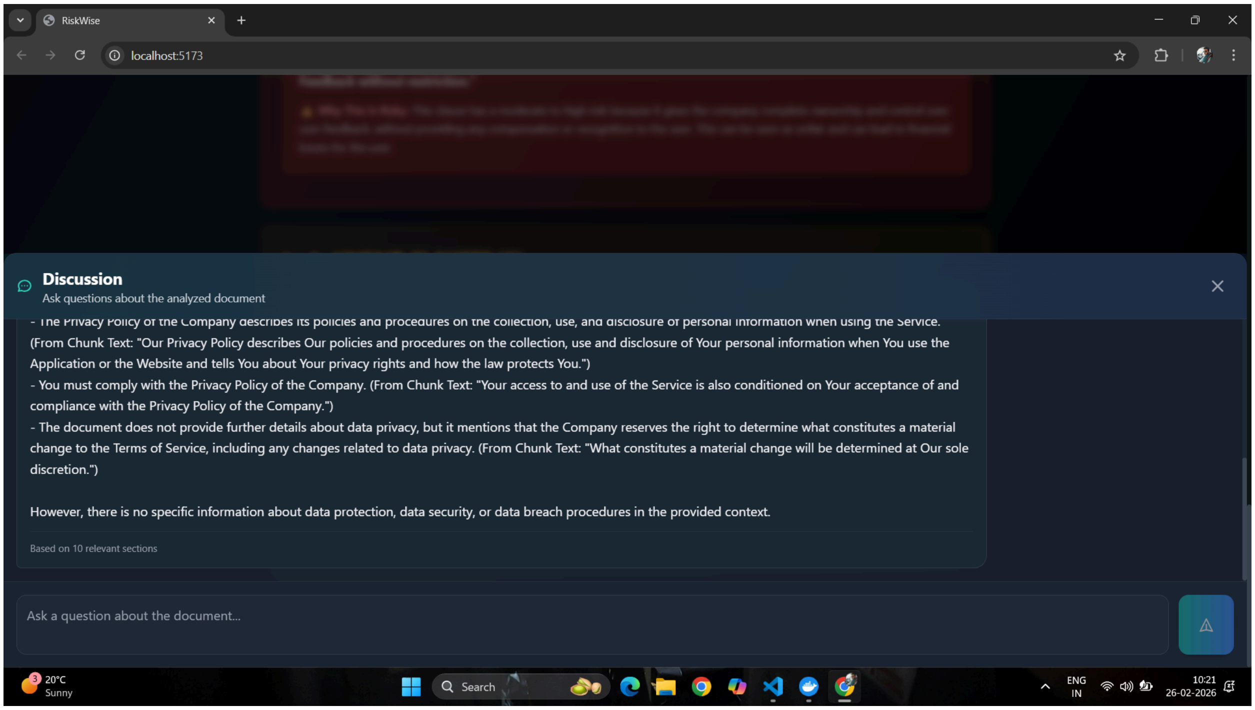

RiskWise
localhost:5173
HIGH RISK CLAUSES (5)
These clauses could seriously harm you. Consider avoiding this service or proceed with extreme caution.
Liability High Risk
Risk: 10/10
"By accessing or using the Service You agree to be bound by these Terms of Service. If You disagree with any part of these Terms of Service then You may not access the Service. You represent that you are over the age of 18. The Company does not permit those under 18 to use the Service."
Why This Is Risky: This clause has a high risk because it creates a blanket liability for users who disagree with the terms of service. If a user disagrees with any part of the terms, they are not allowed to access the service, which can be seen as an unfair restriction. Additionally, the clause requires users to be over 18, which can be problematic for users who are 18 but not yet considered adults in their country of residence.
Liability High Risk
Risk: 9/10
"You are responsible for safeguarding the password that You use to access the Service and for any activities or actions under Your password, whether Your password is with Our Service or a Third-Party Social Media Service. You agree not to disclose Your password to any third party. You must notify Us immediately upon becoming aware of any breach of security or unauthorized use of Your account."
Why This Is Risky: This clause has a high risk because it places the entire responsibility for password security on the user. If a user's password is compromised due to no fault of their own, they may still be held liable for any activities that occur under their
© 2025 RiskWise. A project by Ansul Kumar and Mohammed Amaan Thayyil. This tool provides informational insights, not legal advice.



A.1.3 Conversational Query Interface

The interactive query interface allows users to ask contextual questions regarding identified risks and receive grounded responses based on retrieved document content.



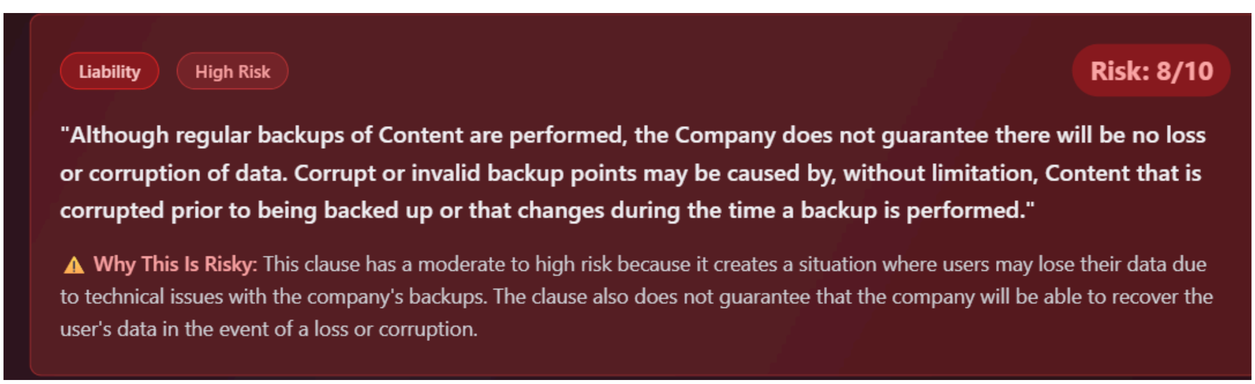


A.2 Sample Analysis Outputs

This section presents representative outputs generated by the RiskWise system to illustrate risk detection and explanation capabilities.

A.2.1 Detected Risk Clause Example

Example of a clause identified as high-risk, including severity score and explanatory reasoning generated by the system.



A.2.2 Contextual Question Answering Example

Example illustrating system response to a user query grounded in retrieved document context.

User: What does the document say about data privacy

Chat: The document mentions data privacy in the following context:

- The Privacy Policy of the Company describes its policies and procedures on the collection, use, and disclosure of personal information when using the Service. (From Chunk Text: "Our Privacy Policy describes Our policies and procedures on the collection, use and disclosure of Your personal information when You use the Application or the Website and tells You about Your privacy rights and how the law protects You.")

- You must comply with the Privacy Policy of the Company. (From Chunk Text: "Your access to and use of the Service is also conditioned on Your acceptance of and compliance with the Privacy Policy of the Company.")

- The document does not provide further details about data privacy, but it mentions that the Company reserves the right to determine what constitutes a material change to the Terms of Service, including any changes related to data privacy. (From Chunk Text: "What constitutes a material change will be determined at Our sole discretion.")

However, there is no specific information about data protection, data security, or data breach procedures in the provided context.

Based on 10 relevant sections

A.3 Prompt Design Examples

The system employs carefully designed prompts for knowledge extraction and risk analysis. Representative prompt templates are provided below.

A.3.1 Triple Extraction Prompt

Prompt guiding language model to extract subject-predicate-object relationships from text.

👉 Paste prompt

You are an information extraction system specialized in Terms of Service.

Your task: From the given text, extract all factual subject–relation–object triples.

Rules:

- Output ONLY in this exact format, one triple per line:

  (SUBJECT, RELATION, OBJECT)

- No explanations, bullet points, numbering, or extra words.

- Keep SUBJECT, RELATION, and OBJECT concise.

- If no valid triples exist, return nothing.

Examples:

Text:

"Marie Curie discovered radium and polonium."

Output:

(Marie Curie, discovered, radium)

(Marie Curie, discovered, polonium)

Text:

"Users cannot share their passwords with anyone."

Output:

(User, cannot_share, passwords)

Text:

"The User must provide accurate information during account registration."

Output:

(User, must_provide, accurate information)

(User, registers_for, Account)

Text:

"The Company may terminate your subscription at any time with notice."

Output:

(Company, may_terminate, subscription)

(Company, gives_notice, User)

Text:

"By using the Service, you agree to the Terms of Service and Privacy Policy."

Output:

(User, agrees_to, Terms of Service)

(User, agrees_to, Privacy Policy)

Text:

"If the User violates the rules, the Company may suspend the account."

Output:

(User, violates, rules)

(Company, may_suspend, account)

Now extract triples from this text:

"{chunk_text}"

A.3.2 Risk Analysis Prompt

Prompt used to instruct the model to identify risky clauses, assign scores, and provide explanations.

You are an expert legal document analyst specializing in Terms of Service agreements.

Your task: Identify the 3-5 MOST RISKY clauses that could harm users or limit their rights.

Instructions:

1. Focus on clauses that create LIABILITY, PRIVACY issues, TERMINATION rights, PAYMENT obligations, or UNFAIR CHANGES

2. Ignore routine legal language - find the CONCERNING clauses

3. For EACH risky clause, provide EXACTLY this format with your reasoning:

CLAUSE: [exact text from the ToS - keep it complete]

RISK: [number 1-10 where: 1-3=minimal risk, 4-6=moderate risk, 7-10=high risk]

CATEGORY: [choose: Privacy, Liability, Termination, Payments, or Changes]

REASON: [2-3 sentences explaining specifically why this is risky for users]

IMPORTANT:

- Leave ONE blank line between each clause

- Risk scores BELOW 7 are NOT truly risky (use 7-10 for actual risks)

- Be specific about WHO is harmed and HOW

Terms of Service:

{document_text}

Now identify the MOST RISKY clauses. Start with the highest risk items.